KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2021
Department of Computer Science and Engineering
CSE 4221
Natural Language Processing

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What are the differences between Natural Language Processing (NLP) and Text Mining? Explain. (06)
   b) What is Lemmatization and Stemming? Discuss one popular Lemmatization and Stemming algorithm with example. (10)
   c) Develop the regular expression to find all the instances of the word "there" in a text. (07)
   d) You have two strings X of length n and Y of length m. Develop an algorithm to edit the string X to Y. How many operations you need? Explain your algorithm for X = "KITTEN" and Y = "SITTIN". (12)

2. a) Define N-gram language model. How can you compute the probability of a sentence of n words P(w), $w = w_1 w_2 \dots w_n$. How can you calculate maximum likelihood by bigram probabilities? (10)
   b) Consider the following corpus. (15)

<S> I Like Henry </S>
<S> I Like College </S>
<S> Do Henry Like College </S>
<S> Henry I am </S>
<S> Do I Like Henry </S>
<S> Do I Like College</S>

What is the most probable next word
   i) "<S> Do —" applying bigram probability model?
   ii) "<S> Do I Like —" applying bigram probability model?
   iii) "<S> Do I Like —" applying trigram probability model?

   c) Define smoothing. How can you smooth the language model by interpolation? How to choose the value of 'λ' to maximize the probability? (10)

3. a) What is Bag of words representation? How do you apply Bag of words assumption and conditional independence for naïve bayes classifier? Explain. (12)
   b) Consider the following Training dataset. (08)

item	class
Just plains boring	–
Entirely Predictable and lacks energy	–
No surprises and very few laughs	–
Very powerful	+
The most fun film of the summer	+

Find the class of the following sentence "Predictable with no fun", using naïve bayes classifier.
   c) How can you apply noisy channel for real world spelling correction? (10)
   d) "Accuracy is not a good metric when the goal is to discover something rare" –justify the statement. (05)

4. a) What are the common structural ambiguities for any NLP task? Give example(s). (08)
   b) Consider the following grammar G (13)

$S \rightarrow NP\ VP$
$NP \rightarrow Det\ N$
$VP \rightarrow V\ NP$
$V \rightarrow includes$
Det → the | a
$N \rightarrow meal \mid flight$

Is the string "The flight includes a meal." in L(G)? Answer it using CYK algorithm.

c) Define PCFG. What is the condition to be maintained by a PCFG? How can you disambiguate the parsing using PCFG? (09)

d) What are the applications of shallow parsing? (05)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is closed class and open class of parts of speech? Explain with example. (08)

b) Give a short description of Viterbi algorithm. What are the applications of the algorithm? (07)

c) Use the following table to answer the following questions. (20)

We have a dataset (NextD). Some information about the dataset is given below.	
No. of VB: 10756	Starting probability VB: 0.44417
No. of MD: 19375	Starting probability MD: 0.7836
No. of NN: 14479	Starting probability NN: 0.25762
$(t_{i-1} - t_i)$	
VB–MD: 9544	VB–race: 6272
VB–NN: 9385	VB–will: 8977
MD–VB: 9918	VB–begin: 5569
MD–NN: 9590	MD–race: 6891
NN–VB: 9065	MD–will: 6301
NN–MD: 9754	MD–begin: 4857
VB–VB: 0	NN–race: 4910
NN–NN: 0	NN–will: 4012
MD–MD: 0	NN–begin: 4617

i) Calculate the transition probability (A).

ii) Calculate the emission probability (B) for sentence: "race will begin".

iii) Find the tag sequence for the sentence "race will begin" using Viterbi algorithm.

6. a) Define Named Entity Recognition. Why recognizing name entity is difficult? (07)

b) Discuss conditional random fields (CRF). Write down the features in a CRF PoS tagger. (08)

c) Discuss the IR-based Factoid Question Answering system with proper diagram(s). (10)

d) Define knowledge-based question answering. Discuss four broad stages of Watson QA with block diagram(s). (10)

7. a) Write down classes of algorithms for relation extraction. Describe Relation Extraction via Supervised Learning and Feature-based supervised relation classifiers. (10)

b) Write down three ways to extract Times and temporal Expression. Explain with examples. (08)

c) Discuss the encoder-decoder model for machine translation with proper figures. (10)

d) Define entity linking. Discuss neural graph-based linking with proper diagram(s). (07)

8. a) Discuss the information Retrieval question-answering system. Explain with examples. (08)

b) Consider the following data: (12)

Query	:	Sweet love
Doc 1	:	Sweet sweet nurse! Love?
Doc 2	:	Sweet sorrow
Doc 3	:	How sweet is love?
Doc 4	:	Nurse!

Using Term Weighting and document scoring, Rank the document for query.

c) What is Homograph disambiguation? What are the problems of CMU? How does UNISYN overcome the problems of CMU? (07)

d) Speech synthesis performs text-to-waveform mapping in two steps. Name and discuss the steps using Hourglass metaphor. (08)